

PLSC 30600

Week 4: More approaches to estimation

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Winter 2026

Strong ignorability and the propensity score: estimation

i	$X_{[1]i}$	$X_{[2]i}$	$p_D(\mathbf{X}_i)$	$Y_i(0)$	$Y_i(1)$	D_i	Y_i
1	A	0	?	0	?	0	0
2	A	0	?	?	1	1	1
3	B	0	?	1	?	0	1
4	B	0	?	?	1	1	1
5	A	1	?	0	?	0	0
6	A	1	?	?	1	1	1
7	B	1	?	1	?	0	1
8	B	1	?	?	0	1	0
9	A	0	?	1	?	0	0
10	B	1	?	?	1	1	1

Probit MLE: notation and log-likelihood (Aronow and Miller 2019,

Section 5.2.5)

- Data: $(Y_i, D_i, \mathbf{X}_i)$ i.i.d. observations of $(Y, D, \mathbf{X})$, $D_i \in \{0, 1\}$, $\mathbf{X}_i \in \mathbb{R}^{K+1}$ includes intercept.
- We want to predict D . Stack outcomes and regressors:

$$\mathbf{D} = \begin{bmatrix} D_1 \\ D_2 \\ \vdots \\ D_n \end{bmatrix}, \mathbf{X}_i = \begin{bmatrix} 1 \\ X_{[1]i} \\ X_{[2]i} \\ \vdots \\ X_{[K]i} \end{bmatrix},$$

and

$$\mathbb{X} = \begin{pmatrix} \mathbf{X}_1^\top \\ \mathbf{X}_2^\top \\ \vdots \\ \mathbf{X}_n^\top \end{pmatrix} = \begin{pmatrix} 1 & X_{[1]1} & X_{[2]1} & \cdots & X_{[K]1} \\ 1 & X_{[1]2} & X_{[2]2} & \cdots & X_{[K]2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{[1]n} & X_{[2]n} & \cdots & X_{[K]n} \end{pmatrix}.$$

Probit MLE: notation and log-likelihood

- D_i is binary, so conditional on $\mathbf{X}_i$, $D_i \sim \text{Bernoulli}(p_i)$, so the likelihood is

$$\mathcal{L}(\mathbf{b} \mid \mathbf{D}, \mathbb{X}) = \prod_{i=1}^n p_i^{D_i} (1 - p_i)^{1-D_i}.$$

- Suppose that

$$\Pr[D = 1 \mid \mathbf{X}] = p(\mathbf{X}; \beta) = \Phi(\mathbf{X}^\top \beta), \text{ where } \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_K \end{bmatrix}.$$

- then the likelihood,

$$\mathcal{L}(\mathbf{b} \mid \mathbf{D}, \mathbb{X}) = \prod_{i=1}^n (\Phi(\mathbf{X}_i^\top \mathbf{b}))^{D_i} (1 - \Phi(\mathbf{X}_i^\top \mathbf{b}))^{1-D_i},$$

log-likelihood,

$$\begin{aligned} \ell(\mathbf{b} \mid \mathbf{D}, \mathbb{X}) &= \log \mathcal{L}(\mathbf{b} \mid \mathbf{D}, \mathbb{X}) \\ &= \sum_{i=1}^n [D_i \log (\Phi(\mathbf{X}_i^\top \mathbf{b})) + (1 - D_i) \log (1 - \Phi(\mathbf{X}_i^\top \mathbf{b}))], \end{aligned}$$

and MLE:

$$\hat{\beta}_{MLE} = \arg \max_{\mathbf{b} \in \mathbb{R}^{K+1}} \ell(\mathbf{b} \mid \mathbf{D}, \mathbb{X}).$$

Probit MLE: estimator and numerical solution

- No closed form; we optimize numerically (Newton / quasi-Newton, e.g., BFGS).
- In code we usually minimize the negative log-likelihood:

$$\hat{\mathbf{b}}_{\text{MLE}} \in \arg \min_{\mathbf{b}} \left(-\ell(\mathbf{b} \mid \mathbf{D}, \mathbb{X}) \right).$$

Code: probit propensity scores (manual MLE)

```
> df <- data.frame(  
+   X_1 = c("A", "A", "B", "B", "A", "A", "B", "B", "A", "B"),  
+   X_2 = c(0,0,0,0,1,1,1,1,0,1),  
+   D    = c(0,1,0,1,0,1,0,1,0,1),  
+   Y    = c(0,1,1,1,0,1,1,0,0,1)  
+ )  
> df$X_1 <- factor(df$X_1)  
> X <- model.matrix(~ X_1 + X_2, data = df)  
> head(X)  
  
      (Intercept) X_1B X_2  
1             1     0   0  
2             1     0   0  
3             1     1   0  
4             1     1   0  
5             1     0   1  
6             1     0   1  
  
> # we are predicting *treatment*  
> D <- df$D  
>
```

Code: probit propensity scores (manual MLE)

```
> neg_loglik <- function(beta, X, D) {  
+   eta <- as.vector(X %*% beta)  
+   p <- pnorm(eta)  
+   -sum(D * log(p) + (1 - D) * log(1 - p))  
+ }  
  
> fit <- optim(  
+   par = rep(0, ncol(X)),  
+   fn = neg_loglik,  
+   X = X,  
+   D = D  
+ )  
  
> round(beta_hat <- fit$par, 3)  
  
[1] -0.431  0.431  0.431  
  
> df$p_hat <- pnorm(as.vector(X %*% beta_hat))  
> round(df$p_hat, 3)  
  
[1] 0.333 0.333 0.500 0.500 0.500 0.500 0.667 0.667 0.333 0.667
```

Code: probit propensity scores (glm)

```
> fit_glm <- glm(D ~ X_1 + X_2, data = df,  
+               family = binomial(link = "probit"))  
> df$p_hat_glm <- predict(fit_glm, type = "response")  
> # are they different?  
> round(cbind(manual = df$p_hat, glm = df$p_hat_glm), 3)
```

	manual	glm
[1,]	0.333	0.333
[2,]	0.333	0.333
[3,]	0.500	0.500
[4,]	0.500	0.500
[5,]	0.500	0.500
[6,]	0.500	0.500
[7,]	0.667	0.667
[8,]	0.667	0.667
[9,]	0.333	0.333
[10,]	0.667	0.667

```
> max(abs(df$p_hat - df$p_hat_glm))
```

```
[1] 6.843044e-05
```


Science table: hot deck imputation (propensity score matching)

i	$X_{[1]i}$	$X_{[2]i}$	$\hat{p}_D(\mathbf{X}_i)$	$Y_i(0)$	$Y_i(1)$	D_i	Y_i
1	A	0	0.33	0	(donor)	0	0
2	A	0	0.33	(donor)	1	1	1
3	B	0	0.50	1	(donor)	0	1
4	B	0	0.50	(donor)	1	1	1
5	A	1	0.50	0	(donor)	0	0
6	A	1	0.50	(donor)	1	1	1
7	B	1	0.67	1	(donor)	0	1
8	B	1	0.67	(donor)	0	1	0
9	A	0	0.33	1	(donor)	0	0
10	B	1	0.67	(donor)	1	1	1

- For each missing potential outcome, choose a nearest-neighbor donor in $p_D(\mathbf{X})$ from the opposite treatment arm.
- Impute $\hat{Y}_i(0)$ or $\hat{Y}_i(1)$ with that donor's observed outcome.

Code: hot deck imputation with hot.deck

```
> library(hot.deck)
> # Impute missing Y(0) for treated units (D=1) using p_hat.
> df$Y0 <- ifelse(df$D == 0, df$Y, NA)
> imp0 <- hot.deck(df[, c("Y0", "p_hat")],
+                 m = 1, method = "p.draw")
> df$Y0_imp <- imp0$data[[1]]$Y0
> # Impute missing Y(1) for control units (D=0) using p_hat.
> df$Y1 <- ifelse(df$D == 1, df$Y, NA)
> imp1 <- hot.deck(df[, c("Y1", "p_hat")],
+                 m = 1, method = "p.draw")
> df$Y1_imp <- imp1$data[[1]]$Y1
> mean(df$Y1_imp - df$Y0_imp)

[1] 0.5
```

IPW: missing data (MAR)

- Reweight observed outcomes by inverse response propensities.
- Theorem (Aronow–Miller 6.2.6): if $Y_i \perp R_i \mid \mathbf{X}_i$, then

$$\mathbb{E}[Y_i] = \mathbb{E}\left[\frac{Y_i^* R_i}{p_R(\mathbf{X}_i)}\right].$$

- Plug-in estimator:

$$\hat{\mathbb{E}}_{IPW}[Y_i] = \frac{1}{n} \sum_{i=1}^n \frac{Y_i^* R_i}{\hat{p}_R(\mathbf{X}_i)}.$$

IPW: causal effects (strong ignorability)

- With $D_i \in \{0, 1\}$ and $p_D(\mathbf{X}_i) = \Pr[D_i = 1 \mid \mathbf{X}_i]$,
- Theorem (Aronow–Miller 7.2.5):

$$\mathbb{E}[\tau_i] = \mathbb{E}\left[\frac{Y_i D_i}{p_D(\mathbf{X}_i)} - \frac{Y_i(1 - D_i)}{1 - p_D(\mathbf{X}_i)}\right].$$

- Plug-in estimator:

$$\hat{\mathbb{E}}_{IPW}[\tau_i] = \frac{1}{n} \sum_{i=1}^n \left(\frac{Y_i D_i}{\hat{p}_D(\mathbf{X}_i)} - \frac{Y_i(1 - D_i)}{1 - \hat{p}_D(\mathbf{X}_i)} \right).$$

IPW: creating a pseudo-population

$X_{[1]}$	$X_{[2]}$	n	$\hat{p}_D(\mathbf{X})$	n_1 (treated)	n_0 (control)
A	0	3	0.33	1	2
A	1	2	0.50	1	1
B	0	2	0.50	1	1
B	1	3	0.67	2	1

$X_{[1]}$	$X_{[2]}$	n	$\hat{p}_D(\mathbf{X})$	$\sum_{i:D_i=1} \frac{1}{\hat{p}_D(\mathbf{x}_i)}$	$\sum_{i:D_i=0} \frac{1}{1-\hat{p}_D(\mathbf{x}_i)}$
A	0	3	0.33	$1/0.33 \approx 3$	$2/0.67 \approx 3$
A	1	2	0.50	$1/0.50 = 2$	$1/0.50 = 2$
B	0	2	0.50	$1/0.50 = 2$	$1/0.50 = 2$
B	1	3	0.67	$2/0.67 \approx 3$	$1/0.33 \approx 3$

- Interpretation: IPW creates a pseudo-population where, within each cell, treated and control have equal weighted mass.

Code: IPW ATE by hand (using $\hat{p}_D$)

```
> ipw_term <- (df$Y * df$D / df$p_hat) -  
+   (df$Y * (1 - df$D) / (1 - df$p_hat))  
> ipw_ate <- mean(ipw_term)  
> ipw_ate  
[1] 0.3499516
```

IPW in matrix notation

$$\mathbb{X} = \begin{bmatrix} \mathbf{X}_1^\top \\ \vdots \\ \mathbf{X}_n^\top \end{bmatrix} \in \mathbb{R}^{n \times k}, \quad \mathbf{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} \in \mathbb{R}^n, \quad \mathbf{D} = \begin{bmatrix} D_1 \\ \vdots \\ D_n \end{bmatrix} \in \{0, 1\}^n, \quad \mathbf{1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^n.$$

Row-wise estimated propensity scores:

$$\hat{\boldsymbol{\rho}} = \hat{\rho}_D(\mathbb{X}) := \begin{bmatrix} \hat{\rho}_D(\mathbf{X}_1) \\ \vdots \\ \hat{\rho}_D(\mathbf{X}_n) \end{bmatrix} \in (0, 1)^n.$$

IPW in matrix notation

Let $\oslash$ denote elementwise division. Define diagonal weight matrices

$$\mathbf{W}_1 = \text{diag}(\mathbf{D} \oslash \hat{\mathbf{p}}), \quad \mathbf{W}_0 = \text{diag}((\mathbf{1} - \mathbf{D}) \oslash (\mathbf{1} - \hat{\mathbf{p}})).$$

i.e., $\text{diag}(\cdot)$ places the weights on the diagonal so each Y_i is reweighted separately.

$$\mathbf{W}_1 = \begin{bmatrix} \frac{D_1}{\hat{p}_D(\mathbf{X}_1)} & 0 & \cdots & 0 \\ 0 & \frac{D_2}{\hat{p}_D(\mathbf{X}_2)} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{D_n}{\hat{p}_D(\mathbf{X}_n)} \end{bmatrix}, \quad \mathbf{W}_0 = \begin{bmatrix} \frac{1-D_1}{1-\hat{p}_D(\mathbf{X}_1)} & 0 & \cdots & 0 \\ 0 & \frac{1-D_2}{1-\hat{p}_D(\mathbf{X}_2)} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1-D_n}{1-\hat{p}_D(\mathbf{X}_n)} \end{bmatrix}.$$

IPW ATE (Horvitz–Thompson form):

$$\hat{\tau}_{\text{IPW}} = \frac{1}{n} \mathbf{1}^\top (\mathbf{W}_1 - \mathbf{W}_0) \mathbf{Y} = \frac{1}{n} \sum_{i=1}^n \left(\frac{D_i Y_i}{\hat{p}_D(\mathbf{X}_i)} - \frac{(1 - D_i) Y_i}{1 - \hat{p}_D(\mathbf{X}_i)} \right).$$

IPW: practical cautions

- Extreme weights when $\hat{p}_D(\mathbf{X}_i)$ is near 0 or 1.
- Overlap diagnostics are essential before weighting.
- IPW variance can be large; stabilized weights can help.

Why adjust in an RCT?

- Randomization targets unbiasedness, not necessarily precision.
- Finite-sample imbalance is common; regression can improve precision.
- Caveat: specification matters for valid inference.

Freedman vs. Lin

- Freedman (2008): naive regression adjustment can worsen precision or SEs. Freedman (2008)
- Lin (2013): fully interacted model + robust SEs restores design-based validity. Lin (2013)

Regression adjustment with interactions

- Treatment indicator D_i , covariates $\mathbf{X}_i$, mean-centered covariates $\tilde{\mathbf{X}}_i$

- Fully interacted model:

$$Y_i = \alpha + \tau D_i + \mathbf{X}_i^\top \boldsymbol{\beta} + (D_i \cdot \tilde{\mathbf{X}}_i)^\top \boldsymbol{\gamma} + \varepsilon_i.$$

- The adjusted ATE is $\hat{\tau}$ from this regression.

Prediction view of the adjusted ATE

- Use the fitted model to predict $\hat{Y}_i(1)$ and $\hat{Y}_i(0)$.
- Then

$$\widehat{ATE} = \frac{1}{n} \sum_{i=1}^n (\hat{Y}_i(1) - \hat{Y}_i(0)).$$

- Equivalent to the treatment coefficient in the interacted model.

Code: estimatr::lm_lin

```
> library(estimatr)
> fit_lin <- lm_lin(Y ~ D, covariates = ~ X_1 + X_2, data = df)
> fit_lin
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.000000e-01	8.750655e-18	5.713858e+16	5.629029e-67
D	3.571429e-01	1.749636e-01	2.041241e+00	1.107872e-01
X_1B_c	1.000000e+00	1.226363e-16	8.154193e+15	1.357148e-63
X_2_c	6.075104e-17	1.310192e-16	4.636803e-01	6.669853e-01
D:X_1B_c	-1.285714e+00	3.499271e-01	-3.674235e+00	2.131164e-02
D:X_2_c	-2.857143e-01	3.499271e-01	-8.164966e-01	4.600508e-01

	CI Upper	DF
(Intercept)	5.000000e-01	4
D	8.429196e-01	4
X_1B_c	1.000000e+00	4
X_2_c	4.245188e-16	4
D:X_1B_c	-3.141609e-01	4
D:X_2_c	6.858391e-01	4

Code: estimatr::lm_lin

```
> library(estimatr)
> lm_0 <- lm_robust(Y ~ X_1 + X_2, data = df[which(df$D == 0), ])
> lm_1 <- lm_robust(Y ~ X_1 + X_2, data = df[which(df$D == 1), ])
> Y0 <- predict(lm_0, newdata = df)
> Y1 <- predict(lm_1, newdata = df)
> mean(Y1 - Y0)

[1] 0.3571429

> fit_lin$coefficients['D']

      D
0.3571429
```

References I

- Aronow, P. M. and Miller, B. T. (2019). *Foundations of agnostic statistics*. Cambridge University Press.
- Freedman, D. A. (2008). On regression adjustments in experiments with several treatments. *The Annals of Applied Statistics*, 2(1):176–196.
- Lin, W. (2013). Agnostic notes on regression adjustments to experimental data: Reexamining freedman's critique. *The Annals of Applied Statistics*, 7(1):295–318.